Autonomous Penetration Trajectory Control for Time-sensitive Targets and Dynamic Threats

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***Abstract*—Aiming at the optimization problem of the unmanned aerial vehicle (UAV) trajectory control in ground penetration combat, the 3-DOF particle model and dynamics model of UAV were established in combination with battlefield environment. Constraints such as initial conditions, terminal conditions and flight performance are considered. Considering the radar cross section (RCS), the dynamic RCS threat model is established. The trajectory optimization model of UAV ground penetration operation under the theoretical framework of optimal control rate was constructed, and the autonomous trajectory control of time-sensitive target under uncertain conditions was solved by combining the receding horizon control strategy. On this basis, taking the flight time and threat probability of UAV to combat area as the objective function, an adaptive weighted performance index and switching sub-objective function method are proposed, and the simulation is carried out. The results show that this method can optimize UAV penetration trajectory control more reasonably against time-sensitive targets in complex battlefield environment and multiple constraints, which is beneficial to realize attack mission.**

Keywords—UAV penetration, trajectory control, receding horizon control, pseudo-spectral method, time-sensitive target, dynamic environment

I. Introduction

In today's battlefield, unmanned aerial vehicle (UAV) is often faced with the task of penetrating into the enemy's complex battlefield environment to carry out ground penetration operations. UAV has become the focus of current weapons and equipment development, and it is urgent for UAV combat mode to adapt to uncertain environmental conditions and meet the needs of precise attack on time-sensitive targets.

In order to realize the ground penetration combat of UAV, most researches use the technique of flight path planning, which is one of the key techniques for successful attack. At present, UAV flight path planning methods mainly include Voronoi graph algorithm [1], A* algorithm [2], SPARSE A* search (SAS) algorithm [3], artificial potential field method [4], gray Wolf algorithm [5], and Rapidly-exploration Random tree (RRT) algorithm [6]. These algorithms have their own advantages, but it is difficult to realize the high-precision control of attack process, and the flightability of planned flight path is also difficult to guarantee. Online flight path planning mainly obtains real-time dynamic flight path through receding horizon control [7]. Research on UAV trajectory planning mainly focuses on static mission scenarios with known local threat information, without considering the dynamic changes of complex battlefield environment and time-sensitive target attack requirements.

Pseudo-spectral method [8] is a proposed method to solve trajectory control optimization problems in recent years. This method can not only obtain multidimensional state information, but also integrate the dynamics and kinematics constraints of UAV into the problem solving framework. Therefore, the practicability and flightability of trajectory are improved a lot. The core idea is to discretize state variables and control variables at specific points, and transform complex differential and integral models into nonlinear programming (NLP) problems by using orthogonal polynomials, and then solve them.

Receding Horizon Control (RHC) [9], also known as ModelPredictive Control (MPC), is a Control method developed in the 1970s. Its core idea is the rolling optimization strategy for the optimal control problem. This method transforms a problem from global optimization solution to local optimization solution in a series of time Windows carried out by online rolling.

Since it is difficult for UAV to carry out reasonable trajectory control for time-sensitive target in uncertain environment, this paper adopts the method of combining receding horizon control strategy with pseudo-spectral method to complete the penetration combat task against time-sensitive target. Therefore, in the process of UAV's ground penetration, the flight trajectory from the initial point to the target point is gradually calculated with time, and the receding horizon control strategy is adopted. Each small section of the trajectory is equivalent to solving a finite time-domain constrained optimization problem. Finally, UAV can realize the penetration trajectory control to adapt to dynamic

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environment and attack time-sensitive targets. In addition, trajectory control optimization must be able to meet the requirements of real-time performance. With the improvement of the processing capacity of scientific computers and various optimization solutions, computing time is no longer a problem.

II. Ground Penetration Model

A. UAV Dynamical Model

UAV needs to control the trajectory of ground penetration process, and the 3-DOF particle model can meet the requirements of trajectory planning. Assuming that the effects of random wind fields are ignored, the model definition is as follows:

$$\begin{cases} \dot{x} = V\cos\theta\cos\varphi \\ \dot{y} = V\cos\theta\sin\varphi \\ \dot{z} = V\sin\theta \\ \dot{V} = g\left(n_x - \sin\theta\right) \\ \dot{\theta} = \dfrac{g\left(n_z\cos\gamma - \cos\theta\right)}{V} \\ \dot{\varphi} = \dfrac{gn_z\sin\gamma}{V\cos\theta} \end{cases} \quad (1)$$

Where, (x, y, z) is the position of UAV in the inertial coordinate system, V is the true airspeed of UAV, θ is the inclination angle, φ is the azimuth angle; n_x, n_z and γ are the control variable of UAV, n_x is tangential acceleration along the velocity direction, n_z is normal overload in the lift direction of UAV, and γ is the roll angle.

$$\begin{cases} n_x = \dfrac{T\cos\alpha - D}{mg} \\ n_z = \dfrac{T\sin\alpha + L}{mg} \end{cases} \quad (2)$$

Where, α is the angle of attack, m is mass of UAV, g is gravitational acceleration, T is thrust, D is resistance, and L is lift.

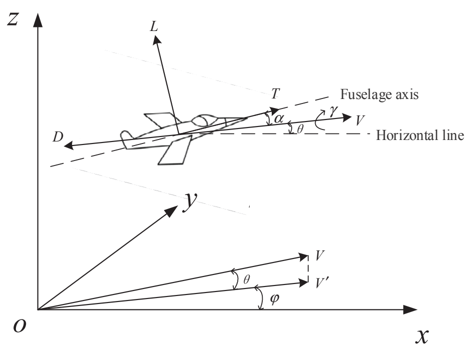


Fig. 1. Diagram of UAV motion dynamics model.

B. Penetration Constraint Model

The initial conditions and the terminal conditions of weapon delivery for UAV trajectory control are as follows:

$$\begin{cases} x(t_0) = x_0, y(t_0) = y_0, z(t_0) = z_0, V(t_0) = V_0 \\ \theta(t_0) = \theta_0, \varphi(t_0) = \varphi_0, \gamma(t_0) = \gamma_0, \alpha(t_0) = \alpha_0 \\ m(t_0) = m_0 \end{cases} \quad (3)$$

Initial constraints, including the position, speed, attitude and mass of the UAV.

$$\begin{cases} \left|x_w - x_f\right| \le \Delta x, \left|y_w - y_f\right| \le \Delta y, \left|z_w - z_f\right| \le \Delta z \\ V(t_f) = V_f, \theta(t_f) = \theta_f, \varphi(t_f) = \varphi_f, \alpha(t_f) = \alpha_f \end{cases} \quad (4)$$

As mentioned above, $\left(x_w, y_w, z_w\right)$ is the weapon placement position of UAV in penetration; $\left(x_f, y_f, z_f\right)$ is the end position of UAV trajectory control; Δx 、 Δy and Δz is the allowed placement error; V_f 、 θ_f 、 φ_f and α_f are the terminal speed and attitude requirements respectively.

In addition, in order to ensure the real flight ability of UAV trajectory control, the flight performance constraints are as follows:

$$\begin{cases} z_{\min} \le z(t) \le z_{\max}, V_{\min} \le V(t) \le V_{\max} \\ \theta_{\min} \le \theta(t) \le \theta_{\max}, \varphi_{\min} \le \varphi(t) \le \varphi_{\max} \\ \gamma_{\min} \le \gamma(t) \le \gamma_{\max}, \alpha_{\min} \le \alpha(t) \le \alpha_{\max} \\ \left|\dot{\gamma}(t)\right| \le \dot{\gamma}_{\max}, \left|n_x(t)\right| \le n_{x\max}, \left|n_z(t)\right| \le n_{z\max} \end{cases} \quad (5)$$

Where, $\dot{\gamma}_{\max}$ is the maximum change rate of trajectory roll angle, $n_{x\max}$ is the maximum tangential overload, and $n_{z\max}$ is the maximum normal overload.

C. Threat Cost Model

In the ground penetration operation, UAV is mainly faced with the enemy's air defense radar, air defense artillery, surface to air missile and enemy aircraft interception and other threats. In order to better describe the success probability of UAV penetration task under the threat of air defense radar, the position and attitude model of UAV relative to air defense radar is established, as shown in Fig.2

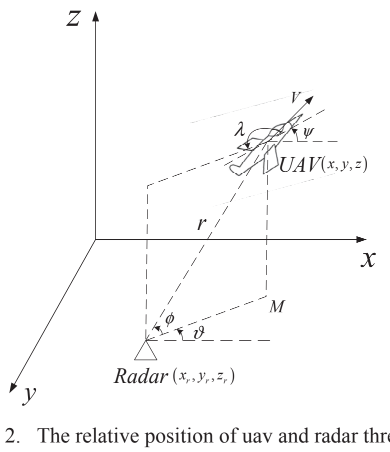


Fig. 2. The relative position of uav and radar threat.

The calculation formula of relevant parameters in Fig.2 is as follows:

$$\begin{cases} x' = x - x_r, y' = y - y_r, z' = z - z_r \\ \vartheta = \arctan(y'/x'), \lambda = \vartheta - \psi + \pi \\ r = \sqrt{x'^2 + y'^2 + z'^2} \\ \phi = \arctan(z / \sqrt{x'^2 + y'^2}) \end{cases} \quad (6)$$

Using the model mentioned in reference [10], the dynamic 3D RCS ellipsoid model is established as follows:

$$\begin{cases} \sigma(\lambda,\phi,\gamma) = \\ \dfrac{\pi a^2 b^2 c^2}{(a^2 \sin^2 \lambda_e \cos^2 \gamma_e + b^2 \sin^2 \lambda_e \sin^2 \gamma_e + c^2 \cos^2 \lambda_e)} \\ \lambda_e = \arccos(\cos\phi \cos\lambda) \\ \gamma_e = \gamma - \arctan(\dfrac{\tan\phi}{\sin\lambda}) \end{cases} \quad (7)$$

Where, σ is the cross-sectional area of UAV radar scattering, a , b and c are the characteristic parameters.

Assume that the probability of instantaneous capture of UAV at a certain moment is as follows:

$$P_t = 1/[1 + (c_2 R_t^4 / \sigma_t)^{c_1}] \quad (8)$$

Where, c_1 and c_2 are the parameters of the radar itself, R_t is the relative distance between the UAV and the radar at the time, and σ_t is the radar scattering cross-sectional area of the UAV at the time.

When the enemy has n air defense radar positions to form a joint air defense system, the detection probability of its air defense system to UAV is as follows:

$$P = 1 - \prod_{i=1}^{n}(1 - P_T(i)) \quad (9)$$

D. Objective Function Model

According to the real-time battlefield situation, the flight time of UAV to the combat area and the threat should be considered comprehensively in most cases. Therefore, the weighted method can be adopted to optimize the objective function:

$$J = \omega_1 J_{time} + \omega_2 J_{threat} = \omega_1 (t_f - t_0) + \omega_2 \int_{t_0}^{t_f} P(X(t))dt \quad (10)$$

Where, ω_1 and ω_2 are the weighting coefficients.

As the above weighting coefficients need some expert experience to adjust and have subjective attributes, they cannot be adjusted reasonably according to the real-time position of UAV. Therefore, this paper proposes an adaptive method to change the weighting coefficient to optimize the objective function for the whole process of UAV's ground penetration, as shown below:

$$J = \begin{cases} J_{time} & P(t) \le 0.3 \\ (1 - \dfrac{r(t)}{v(t)} \cdot \dfrac{1}{t_f - t_0}) J_{time} + \dfrac{r(t)}{v(t)} \cdot \dfrac{1}{t_f - t_0} J_{threat} & 0.3 < P(t) < 0.7 \\ J_{threat} & P(t) \ge 0.7 \end{cases} \quad (11)$$

Where, $r(t)$ is the distance between the position of the UAV and the reference point of the attack target, and $v(t)$ is the approach speed of the UAV and the reference point of the attack target at this moment.

III. Trajectory Control Optimization Solution Framework

In view of the optimization problem of UAV trajectory control for ground penetration, there are many constraints, complex coupling relationship among variables, and nonlinear solutions exist in related control variables and state variables. In order to avoid runge phenomenon, this paper will adopt hp-adaptive-Radau pseudo-spectral method to solve the optimal control problem, and combine the receding horizon control strategy to improve the ability of UAV to adapt to dynamic environment and attack time-sensitive targets.

A. Task Framework Based on Receding Horizon Control Strategy

The receding horizon control strategy generally consists of three basic parts: model prediction, rolling optimization and feedback correction.

1) Model Prediction

Under the framework of the whole task, there are a series of overlapping optimization intervals. The function of model prediction is to predict the future state or output of the system in each interval by combining the relevant information of the system history and a period of input at the future time. According to the real-time environment information, threat and target position information, and UAV its own motion, there is no chance in the control window starting state to estimate the threat environment, a planning window under the target to attack area and UAV flight path, etc., within the window for the next planning trajectory optimization control to lay the foundation.

2) Rolling Optimization

Rolling optimization is the main characteristic of receding horizon control strategy, and it is also the biggest difference from optimal control. According to the previous can optimize the environment of the threat of a window, the target area and UAV flight path information, such as updating the constraint conditions of the next moment, the objective function and related state, this section of pseudo-spectral method to solve the control problem of trajectory, and enforce the results, and push forward the whole rolling time window.

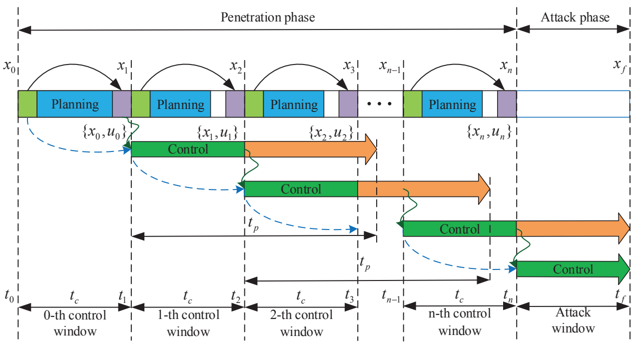


Fig. 3. Trajectory control schematic based on receding horizon control strategy.

Similar to the global optimal trajectory control optimization problem, the optimal trajectory control optimization problem of the first local planning control window can be described as:

$$\begin{cases} \min\ J_k(U)=\int_{kT_c}^{kT_c+Tp} g(X(t),U(t),t)dt+\varphi(X(kT_c+T_p),kT_c+T_p) \\ s.t.\ \dot{X}(t)=f[X(t),U(t),t],t\in[kT_c,kT_c+T_p] \\ \quad \mathrm{C}_k(X(t),U(t),t)\le 0 \\ \quad X(kT_c)=\bar{X}_k \end{cases} \quad (12)$$

Where, $X(t)$ is the state variable, $U(t)$ is the control variable, $\varphi(X(t),t)$ is the residual cost heuristic function, C_k is the updated environmental constraint, and $\bar{X}_k$ is the state feedback obtained by sensor sampling.

3) Feedback Correction

The feedback correction link is a re-adjustment after the implementation of the previous control and planning results, and optimizes the deviation between the actual flight trajectory and the planned trajectory. UAV will encounter complex and changeable situations in the whole penetration attack mission. Due to the imperfect model or the influence of various interference factors, the information in the environment may have changed, especially when the sudden threat is detected and the position of the target is changed after maneuvering. At this time, there is a certain deviation between the real flight state of UAV and the planned state. The above deviation can be transmitted to the model prediction part through feedback correction, and relevant information can be compensated, and then used to optimize the next time window.

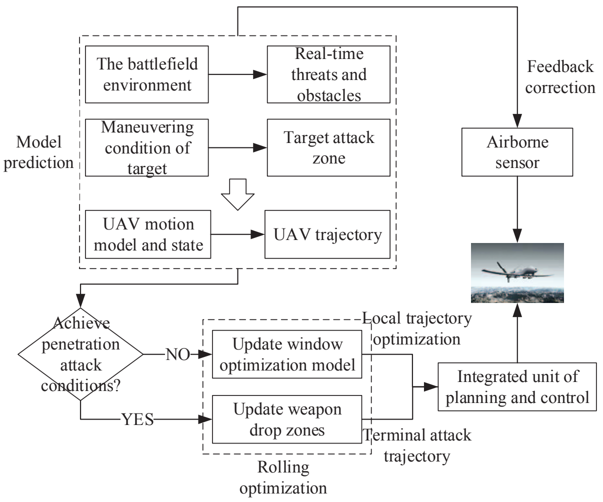


Fig. 4. System structure diagram based on receding horizon control strategy.

B. Task-oriented Adaptive Objective Function

Because the receding horizon control strategy transforms the global optimization problem into multiple continuous sub-optimization problems, its prediction window is limited in time, and it cannot solve the penetration trajectory satisfying the mission requirements within a prediction window.

$$J_{guide}=\frac{r(kT_c+T_p)}{\left|v_c(kT_c+T_p)\right|} \quad (13)$$

Where, $r(kT_c+T_p)$ is the distance between the position of the UAV and the reference point of the attacking target at the terminal moment of the kth rolling optimization window. $v_c(kT_c+T_p)$ is the approach speed of the UAV and the reference point of the attacking target at this moment, the sub-objective function can estimate the attack time of the UAV.

In the process of performing penetration combat tasks, if the UAV knows the threat information in advance, it can conduct trajectory control to avoid the threat in the next optimization window by updating constraint conditions.

$$J_{avoid}=-\int_{kT_c}^{kT_c+T_p}\left(\frac{x(t)-x_{threat}}{a}+\frac{y(t)-y_{threat}}{b}+\frac{z(t)-z_{threat}}{c}\right)dt \quad (14)$$

Where, $(x(t),y(t),z(t))$ is the position of UAV at the current moment, $(x_{threat},y_{threat},z_{threat})$ is the position of emergent threat, a 、 b and c are respectively the safe distance from emergent threat in three coordinate directions.

C. The Solution Strategy of Hp-adaptive Radau Pseudo-spectral Method

The basic idea of pseudo-spectral method is to obtain a series of discrete points by using the linear combination of Legendre polynomials and their first derivatives. At the same time, the time axis is reduced to the corresponding number of points, and state variables and control variables are discretized on these points. The differential equations, constraints and objective functions of the system can be expressed as algebraic equality or algebraic inequality constraints through the unknown states and control variables at these discrete points, and then transformed into NLP. Then, the solution obtained is used as the approximate solution of the original problem by NLP solving algorithm. The transformation process is shown in Fig.5:

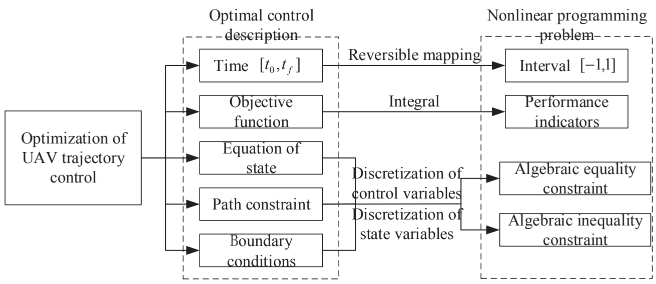


Fig. 5. The transformation of Pseudospectral.

The main steps for constructing NLP problems can be found in reference [11].

The hp-adaptive-Radau pseudo-spectral method can automatically adjust the number of mesh and the order of polynomial according to the curvature of mesh and the error of constraint equation, so it can improve the accuracy of the algorithm with the lowest computational efficiency by combining the advantages of increasing the order and optimizing the number of mesh. The algorithm optimization process includes error criterion, curvature ratio criterion and polynomial order criterion. For specific steps, please refer to reference [12].

IV. Simulation and Discussion

On the basis of the above content, when UAV conducts penetration combat against time-sensitive targets in uncertain environment, the method combining pseudo-spectral method and receding horizon control strategy will be adopted, and a variety of combat scenarios will be set up to verify the effectiveness of this method by simulation.

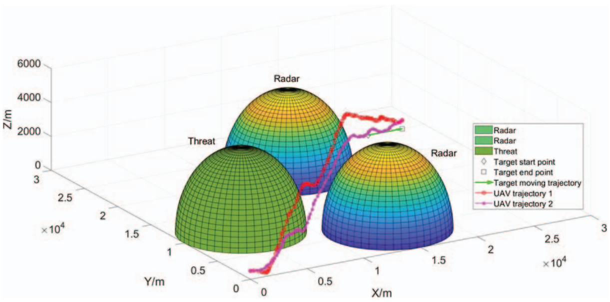
A. Mission Scenario

The combat area is located within the horizontal plane $30km\times 30km$ and the high spatial range $[0,5000]m$ .The initial position of the UAV is $(0,0,500)m$, the initial speed is 204 m/s, the track inclination angle is $\theta = 0^{\circ}$, and the track azimuth angle is $\varphi = 0^{\circ}$; Tangential overload limit is $|n_x(t)| \le 2g$, normal overload limit is $|n_z(t)| \le 4g$, roll Angle limit is $|\gamma(t)| \le 50^{\circ}$; When dropping weapons, UAV speed is not less than 100 m/s, the track inclination angle is $\theta_f = 0^{\circ}$, and the track azimuth angle is $\varphi_f = 40^{\circ}$ .The battlefield known environmental information is that there is an anti-aircraft gun position in the target area at $(5000,1000,0)m$, and two joint alert radars at $(15000,5000,0)m$ and $(15000,20000,0)m$, and all anti-aircraft units have a range of 5 km. In the process of ground penetration, the target will maneuver to a certain extent, and the final condition for the UAV to attack the target is that the distance from the target is less than 500 m, that is, the entire trajectory control optimization is stopped. The detection parameters of the warning radar are $c_1 = 1.01$, $c_2 = 1.25\times10^{-18}$, and the maximum operating distance of the radar position to the target with RCS 1 m^2 is 30 km under ideal conditions. The parameters of the RCS model are $a = 0.3172$, $b = 0.1784$ and $c = 1.003$.

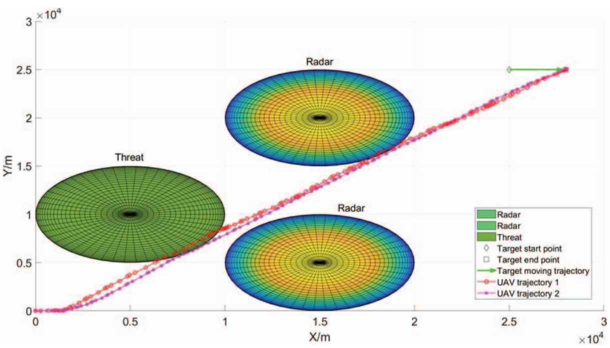
B. Simulation Results and Analysis

Scenario 1: Considering that the target will make a certain maneuver, the initial position of the target is set as $(25000,25000,0)m$, and the target moves uniformly in a straight line along the positive direction of the X-axis at a speed of 15 m/s.

Scenario 2: Under the above simulation conditions, it is assumed that the target movement speed is accelerated to 20 m/s and the terminal time constraint is added, so the UAV needs to attack the ground target at 150 s.



(a) Three-dimensional trajectory diagram.



(b) Trajectory top view.

Fig. 6. For time-sensitive target UAV penetration trajectory.

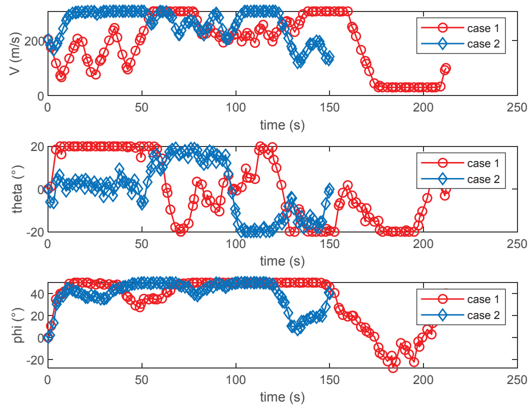


Fig. 7. UAV state variable curve.

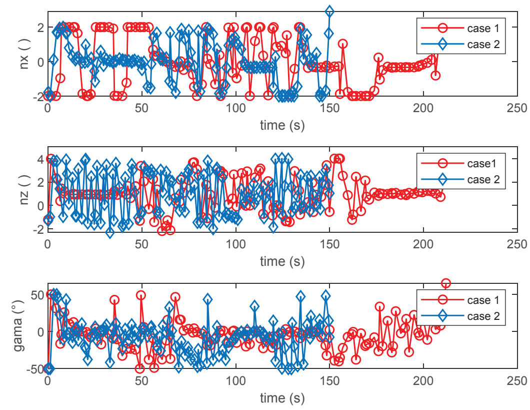


Fig. 8. UAV control variable curve.

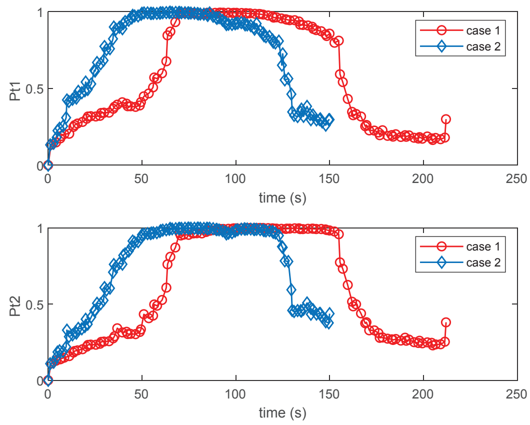
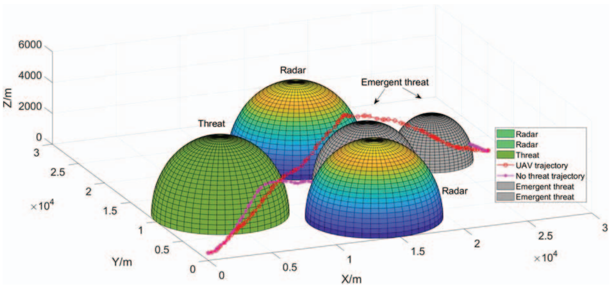


Fig. 9. Warning radar detection probability.

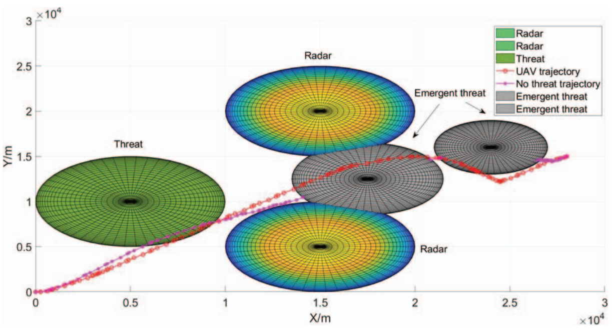
As can be seen from Fig.6 (a) and (b), UAV can optimize trajectory control and complete penetration combat tasks in the case of simple maneuvering of the target. Whether for time-sensitive target or non-time-sensitive target, the trajectory trend of the two cases is basically the same. When the UAV is far from the target, the overall movement of the UAV will not be greatly affected by the target movement. However, when the target is close to a certain distance, the influence of the target position change will be enhanced, and the UAV flight trajectory needs to be constantly adjusted, which is reflected in the trajectory fluctuation. The state and control curves of UAV are shown in Fig.7 and Fig.8

respectively. The state changes in the two cases are relatively smooth. In terms of control quantity, time-sensitive targets change more frequently than non-time-sensitive targets, because the requirements of low detection and fast penetration need to be taken into account in a short time.UAV in each match points are combined warning radar detection probability, as shown in Fig.9, you can see UAV is mainly through the altitude and adjust the attitude to reduce the probability of being detected, but in the middle of the joint through the two radar detection probability is higher, explains the effect will be a discount when both multiple radar. For destroying time-sensitive target, because the overall penetration time limit of 150 s, resulting in the height of the UAV and attitude change is small, see from Fig.6 (a), the UAV can't climb down again or change the posture, the time consuming many times strategy to reduce their RCS, it will affect the whole process of radar detection probability. However, there are advantages and disadvantages. Under the premise of sacrificing low detectability in the process of penetration combat, UAV can reach the corresponding position in effective time and complete the attack on time-sensitive targets. If it is used in multi-aircraft cooperative operations, it can be understood as the need to attack specific targets within the corresponding time window.

Scenario 3: considering the sudden short-range threat of UAV's ground penetration, in the process of UAV's ground penetration, there will be a ground-to-air missile position at the coordinates $(17500,12500,0)m$, and an unexpected threat with an action range of $4\ km$. Then another surface-to-air missile site appeared at coordinates $(24000,16000,0)m$, with a sudden threat of $3\ km$ in scope.



(a) Three-dimensional trajectory diagram.

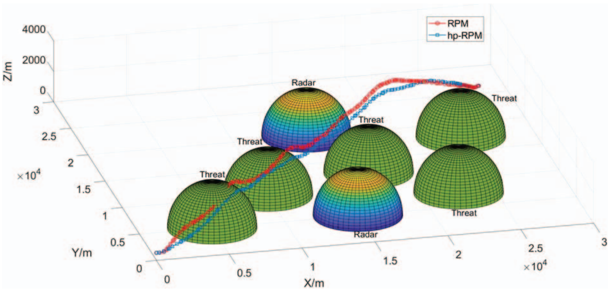


(b) Trajectory top view.

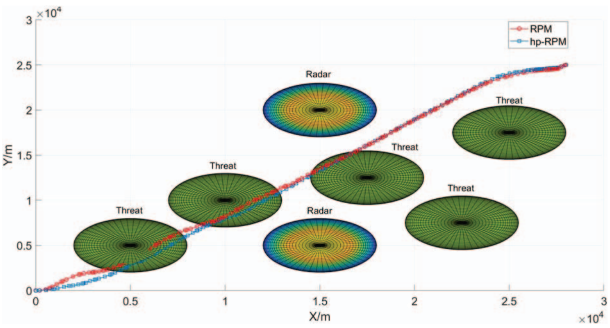
Fig. 10. For sudden threat UAV penetration trajectory.

As can be seen from Fig.10, when there is no emergent threat, the trajectory of UAV is shown as purple curve. In case of emergent threat, the trajectory of UAV is shown in red curve. Compared with the red curve, it can be found that the UAV in the purple curve can reach the destination faster, but through the area of emergent threat. Therefore, if there is a sudden threat, the trajectory control of purple curve is extremely unreasonable, and the UAV is likely to be destroyed by the enemy on the battlefield, resulting in mission failure. When the UAV encounters the first emergent threat, it quickly climbs high to avoid the threat, instead of bypassing because it needs to keep a certain distance from the radar due to its low detecibility. However, when the UAV encounters the second threat, it is already in the altitude decline stage. If the method of altitude climbing continues, it is likely to collide with the threat in the process of altitude adjustment. And the subsequent stage is far from the radar, so the surround threat method is adopted.

Scenario 4: The radar position remains unchanged, the number of threats is increased, the action range is set to 3km, and the final target position is $(28000,25000,500)m$. The general Radau pseudo-spectral method (RPM) and adaptive Radau pseudo-spectral method (hp-RPM) are used for experimental testing. The simulation test results of the two algorithms are shown as follows.



(a) Three-dimensional trajectory diagram.



(b) Trajectory top view.

Fig. 11. Test results of different algorithms.

According to the test results, in algorithm 1 (RPM), the flight time of UAV is $209.85\,s$, and the simulation running time is $20.24\,s$. In algorithm 2 (hp-RPM), the flight time of UAV is $174.46\,s$, and the simulation running time is $14.24\,s$. In contrast, Algorithm 2 computes faster and the drone can reach the target faster. It can be seen from Fig.11 that the UAV in algorithm 1 enters the attack range when avoiding the first threat. However, when the constraints are satisfied, the overall trajectory of the UAV in Algorithm 2 is closer to the threat, with less fluctuation and shorter distance.

V. Conclusion

In this paper, the optimization of UAV trajectory control for ground penetration under uncertain environment is studied, and experimental data of autonomous trajectory control for time-sensitive targets and dynamic threats are analyzed. Taking UAV as the research object, the dynamics and kinematics models of UAV are established. The threat cost function considering the threat blind area and UAV dynamic RCS characteristics was established, and the UAV penetration trajectory control optimization model under uncertain environment was constructed. A solving framework based on the combination of receding horizon control strategy and hp-adaptive Radau pseudo-spectral method was proposed to improve the objective function and meet the actual penetration requirements. The simulation results show that the proposed method can comprehensively consider UAV platform performance, air defense firepower or threat detection, guided weapon projection area and other complex constraints, and generate realistic and feasible optimal penetration trajectory with high accuracy and speed satisfying terminal position, velocity and attitude constraints. And it can reduce RCS by maneuvering to change the attitude angle of the fuselage relative to the radar incident wave to reduce the detection probability to avoid threats, and finally enter the reference point of weapon delivery accurately with the specified attitude and speed.

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